Internal Assessment 02

Full Stack Web Development - Question Paper

Name: _____ Roll No: _____ Date: _____

Max Marks: 50 | Duration: 3 hours | Units: Week 1 to Week 5 (complete full stack application)

Instructions

- Build a working full stack application with Express, MongoDB and Mongoose.
- Keep secrets in a .env file and provide a .env.example.
- Capture a screenshot of the running application as proof.

Problem: Savings Goals Tracker

Build an application where a user can create savings goals, add money to them and see overall progress.

Q1. Data model (10 marks)

Create a Goal model with name (required, max 60 characters), target (required, at least 1), saved (default 0, never negative) and an optional deadline. Add a virtual field progress that returns the percentage completed and never exceeds 100.

Q2. REST API (15 marks)

Implement: GET /api/goals, POST /api/goals, PATCH /api/goals/:id/deposit (adds an amount using an atomic update) and DELETE /api/goals/:id. Return 400 for invalid input or ids and 404 when the goal does not exist.

Q3. Aggregation (10 marks)

Implement GET /api/stats using an aggregation pipeline that returns the number of goals, the total target and the total saved.

Q4. Frontend (10 marks)

Create a page that lists goals with a progress bar, a form to add a goal, and buttons to deposit and delete. The page must refresh its data after every change.

Q5. Documentation (5 marks)

Provide run instructions and the screenshot of the working application.

Deliverables

- src/ (server.js, models/Goal.js, public/index.html, package.json, .env.example)
- screenshots/fullstack-app-demo.png